



GTC4Lusso USA - Seat belts - Latest Update: 2021-01-26

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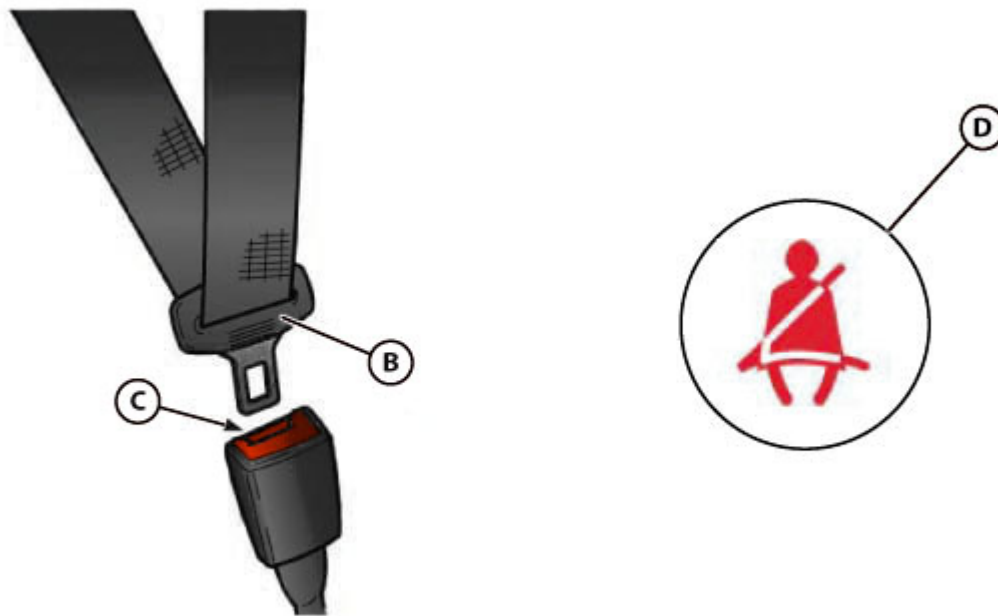
F4.03 Seat belts

Seat belts



The seat belts **(A)** are three-point automatic belts with an inertial locking retractor with pretensioner.

Seat belt fastening



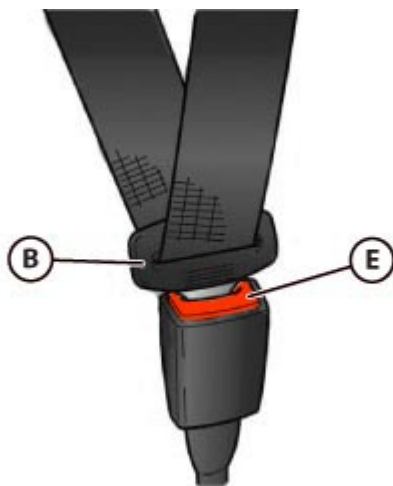
After adjusting the seat correctly, grip the clip (B), pull the belt slowly and insert the tongue into the buckle (C). If the belt locks while pulling, let it rewind a little and then pull it again without jerking abruptly.

Ensure that the tongue clicks audibly into the buckle.

Position the seat belt correctly.

If the driver or passenger seat belts are not buckled, the warning lamp D lights when the ignition key is turned to (II).

Releasing the seat belts



Press the release button (E).

Accompany the buckle (B) back to its rest position.

Pretensioners

This device consists of a retractor with a pyrotechnic pretensioner. The pyrotechnic system is integrated in a single component with the retractor.

This device maximises the protective action of the seat belt.

The pretensioner is activated by the airbag control ECU in the event of a frontal collision (direction of impact between 11 a.m. and 1 p.m.) of excessive severity, or in the event of lateral collision of sufficient severity.

The seat belt is rewound by a few centimetres immediately before the restraining action begins, taking up any slack between the belt and the body of the occupant.



Effective activation of the pretensioner is indicated by the light on the instrument panel and by the tightening of the belt.

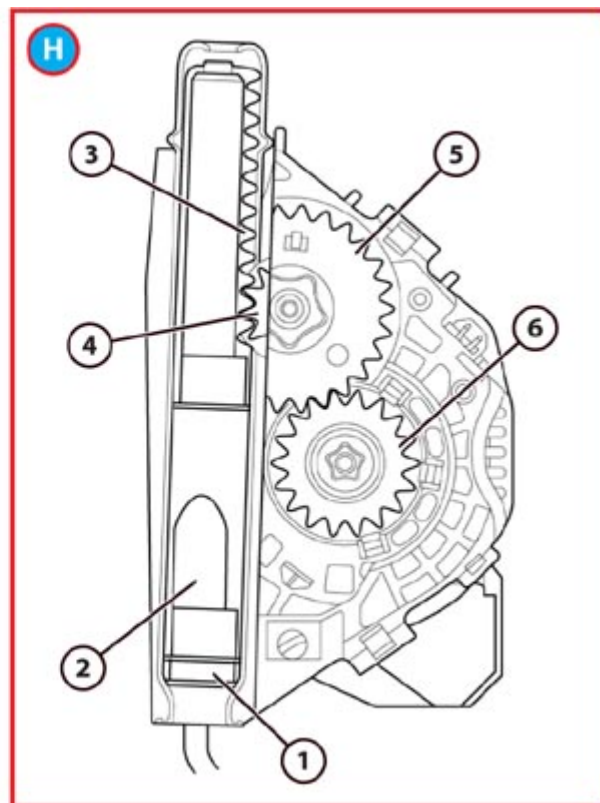
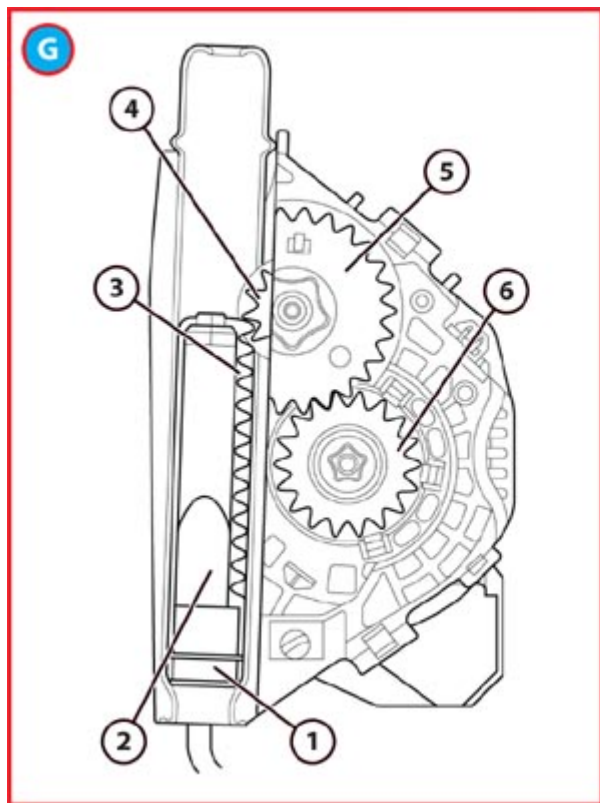
Pretensioner activation releases a small quantity of powder. This powder is harmless and is not an indication of fire.

Pretensioner activation depends only on the state of the seat belt and is not influenced by the presence of the occupant.

If the belt is not buckled, the pretensioner is not activated even if the seat is occupied.

The seat belt is equipped with a load limiter to reduce the restraint force exerted by the belt on the body of the occupant in the event of a collision. This device controls the force with which the seat belt is rewound during a collision.

Operation



Before activation, the device is in **(G)** state.

When the ECU detects a collision of a severity above the set threshold, it sends a signal to the pretensioner, firing the pyrotechnic charge **(1)** of the gas generator: the combustion of the propellant **(2)** releases high pressure gas, exerting an upward force on the rack piston **(3)**.

This motion is transmitted via pinions **(4)** and **(5)** to the toothed bushing **(6)** which rewinds the belt by a few centimetres (state **H**).

This takes up any slack between the belt and the body of the occupant immediately before the retention action begins.



Once activated, a pretensioner is no longer functional, cannot be repaired and must be replaced.

Seat belt and pretensioner care

- Replace the seat belt after a severe collision even if it does not appear damaged.
- Periodically check that the anchor screws are tightly fastened and that the belt itself is in good condition and winds and unwinds smoothly.
- The belt must be kept clean; dirt may compromise the effectiveness of the retractor.
- To clean the belt, hand wash with water and mild soap, rinse and allow to dry. Do not use strong detergents, bleach or harsh solvents that could weaken the fibres.
- Do not wet the retractors: water infiltration may compromise operation.
- The pretensioner requires no maintenance or lubrication. If the pretensioner has been immersed in water or mud, it must be replaced.
- The pretensioners must be replaced every **15 years**.
- Never dismantle or modify the belts, retractors and pretensioners.
- Particular repairs and maintenance work causing violent knocks, vibration or heating near the pretensioner may accidentally trigger the pretensioner; road vibration and shocks will not trigger the pretensioners.

Safety devices

- The pretensioners must be protected from exposure to temperatures of over **100 °C** for more than **6 hours** and from sparks and naked flames.
- The pretensioner may fire spontaneously at temperatures above **150 °C**.
- Pretensioners that have sustained knocks or falls must not be used and must be returned to the manufacturer, specifying the reason.
- Gas generators must be kept away from acids, water, grease or heavy metals to prevent the formation of toxic gases or explosive compounds.
- Do not store pretensioners together with inflammable and/or combustible materials.
- Never dismantle retention system components.
- Never carry pretensioners by the belt, as this may cause damage.
- Remove the pretensioners before carrying out repair jobs that may involve violent shocks (e.g. bodywork/chassis repairs) and/or high temperatures (e.g. painting with temperatures above **100 °C**).
- Fire pretensioners before scrapping.
- Avoid skin contact with the propellant or swallowing: wear protective gloves when handling fired pretensioners. In the event of contact with the propellant, wash with plenty of water and seek medical assistance.

Diagnosing seat belt extender malfunction

BELT GIVER MULFUNCTION CHECKLIST	
Clarify the malfunction: the belt giver is stucked ☐ is noisy ☐ works randomly ☐	
If the belt giver is stucked, the mulfunction is temporary ☐ permanent ☐	
If the belt giver is temporary stucked, clarify if in extension ☐ in return ☐	
If the belt giver is permanently stucked, clarify if in extension ☐ in return ☐	
In case of mulfunction, the belt giver is totally extended ☐ totally back ☐ partially extended (indicate the distance from the trim.....) ☐	
Specify the temperature at which the defect occurs:	
The Warning Light is on YES ☐ NO ☐	
There are errors YES (DTC list requested) ☐ NO ☐	
The belt giver works again if moved manually YES ☐ NO ☐	
Does the beltgiver work again after a battery disconnection? YES ☐ NO ☐	

In all cases of seat belt extender malfunction, please compile the following check list before starting any work.

Replacing the front seat belts

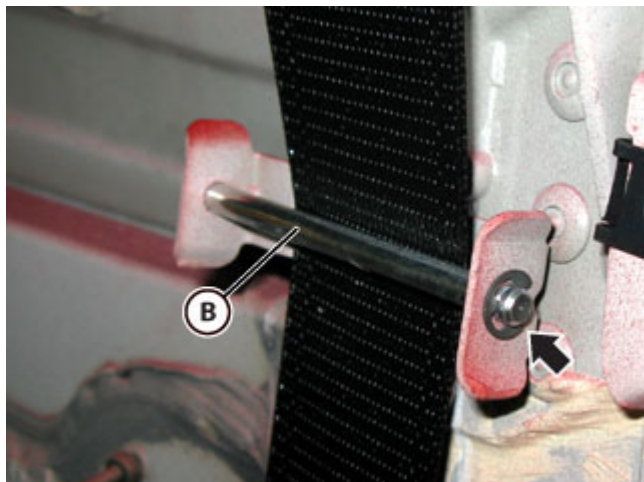
			
Tightening torque		Nm	Class
Fastening the front seat belts	Screw	40 Nm	A
Fastening the front belt winder cover	Screw	4 Nm	B

- Disconnect the battery ( F2.01).
- Remove the roof trim panels ( E4.04).
- Remove the upper B-pillar trim panels.
- Remove the passenger compartment trim panels ( E4.04).
- Remove the lateral rear passenger compartment trim panels.

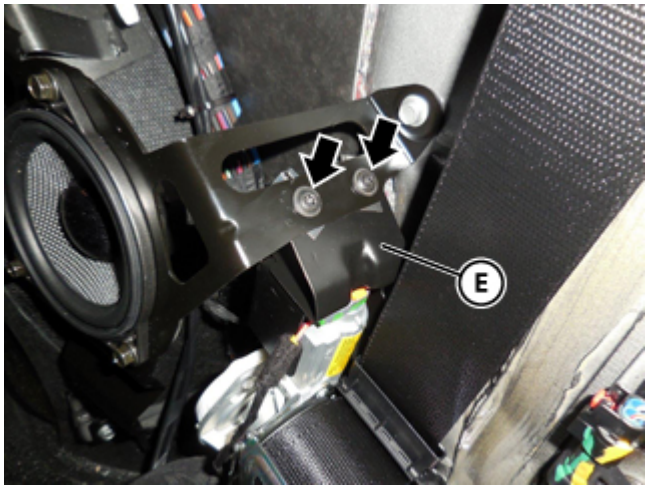
Driver seat belt



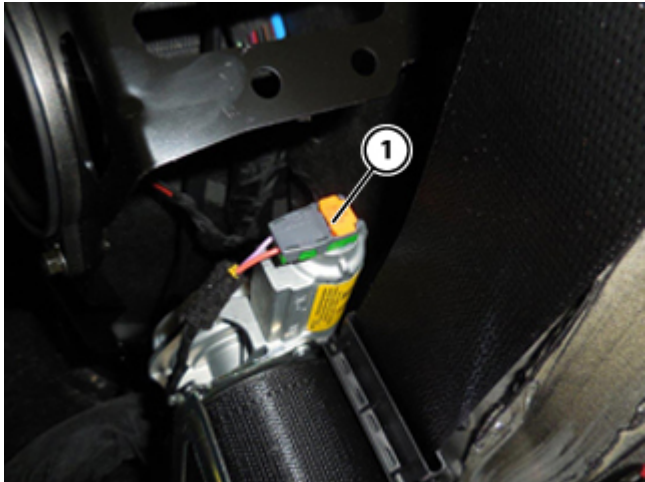
- Undo the indicated screw.
- Remove the turning loop (A).



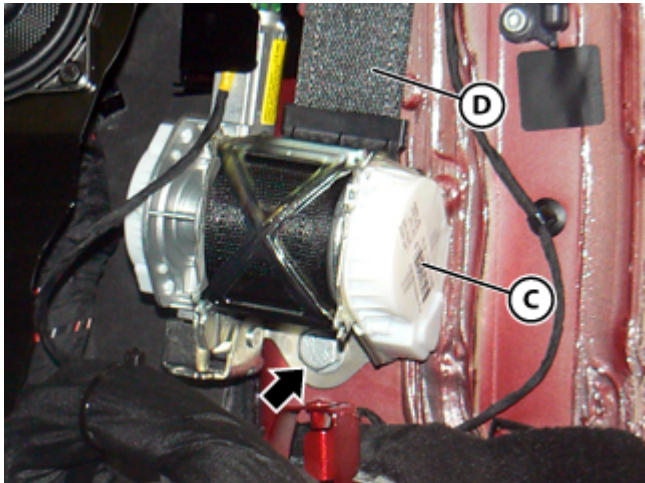
- Remove the indicated circlip.
- Remove the pin (B).



- Undo the screws indicated.
- Remove the cover **(E)**.

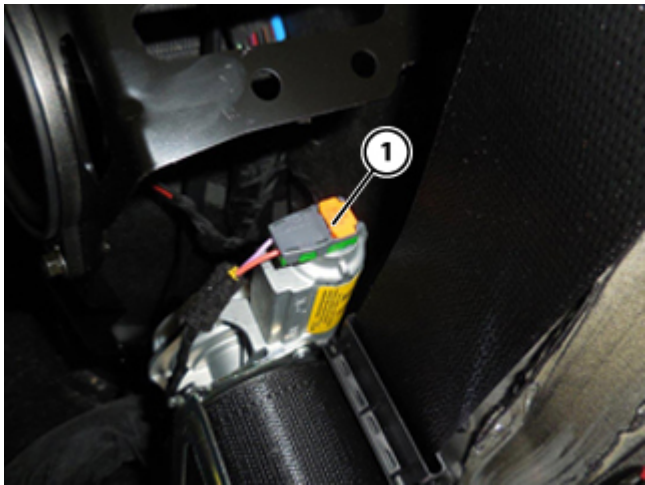


- Disconnect the connector **(1)**.

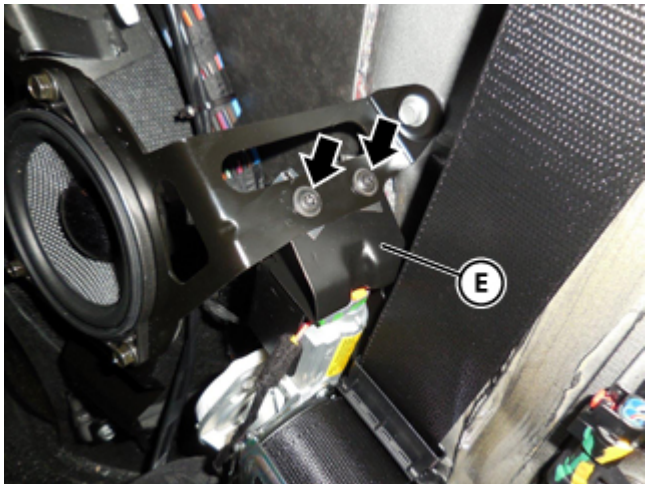


- Undo the indicated screw.
- Remove the belt retractor **(C)** complete with seat belt **(D)**, and replace both.
- Fit the new belt retractor **(C)** complete with seat belt **(D)**.
- Tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A

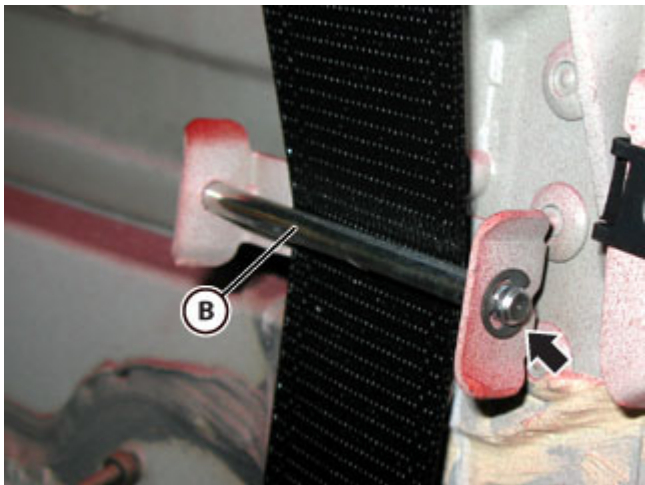


- Connect the connector (1).



- Fit the cover (E).
- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	4 Nm	B



- Fit the pin (B).
- Refit the indicated circlip.



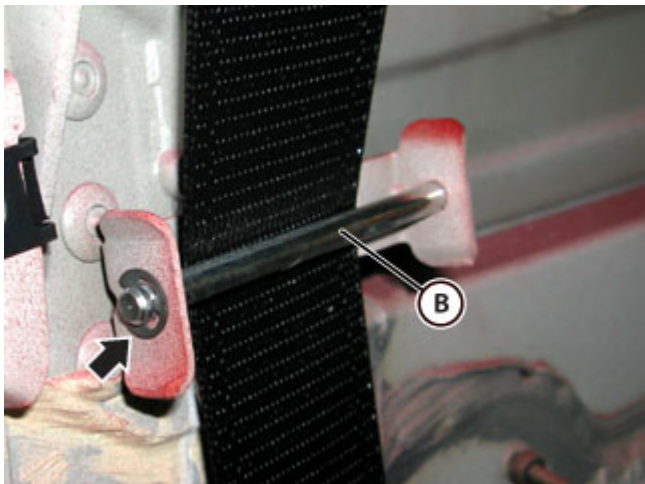
- Fit the turning loop (A).
- Tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A

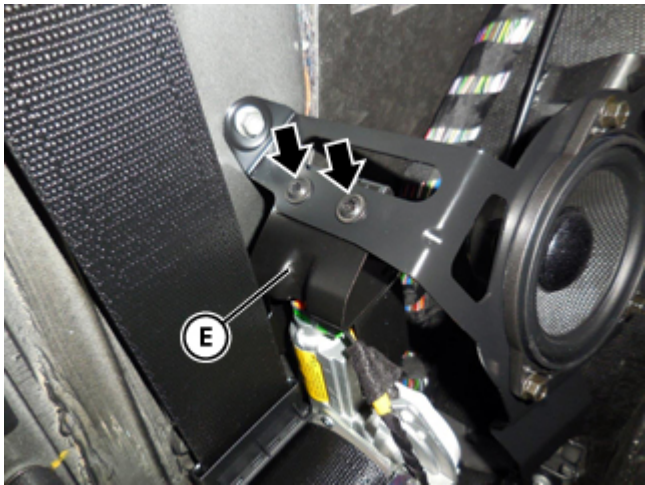
Passenger seat belt



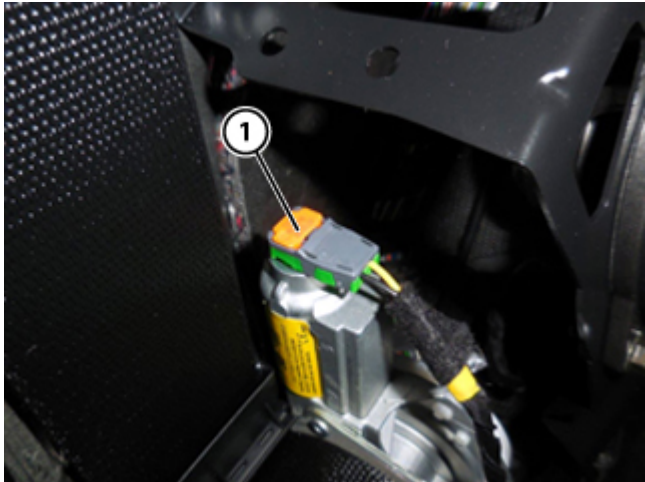
- Undo the indicated screw.
- Remove the turning loop (A).



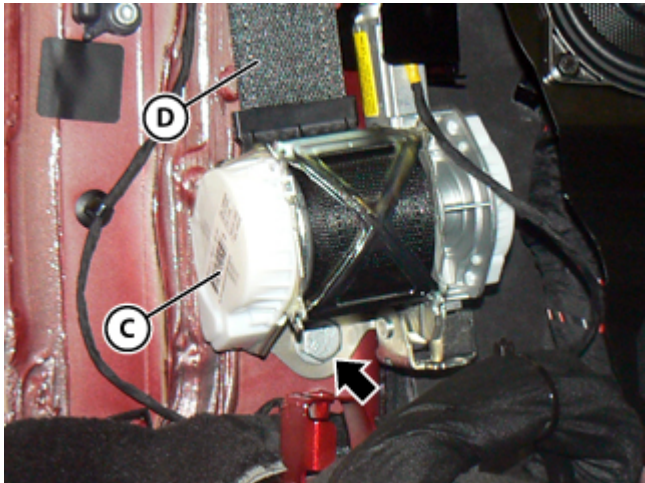
- Remove the indicated circlip.
- Remove the pin (B).



- Undo the screws indicated.
- Remove the cover **(E)**.

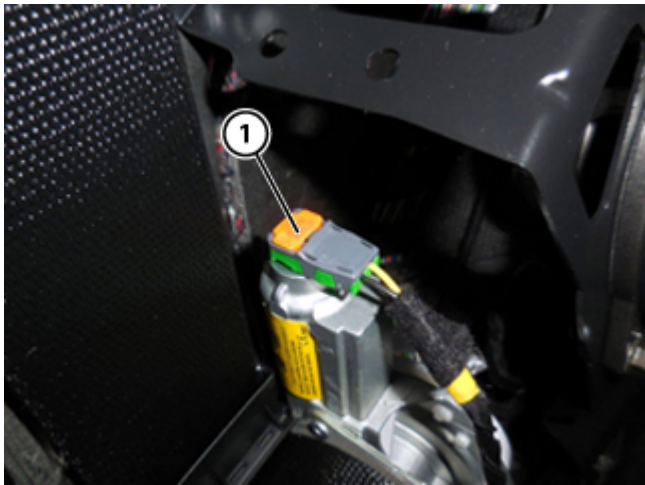


- Disconnect the connector **(1)**.

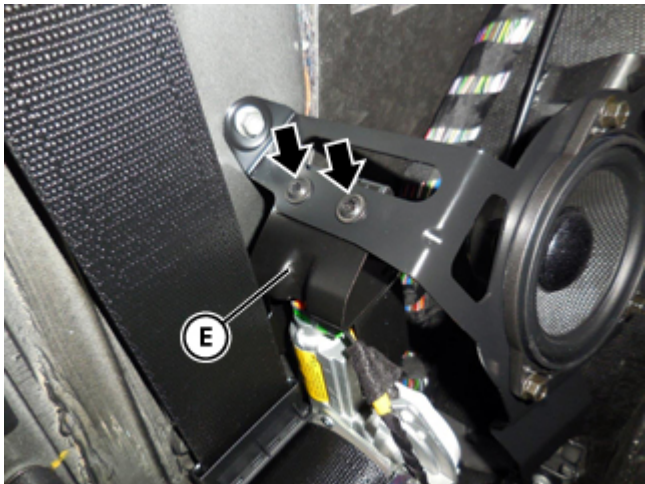


- Undo the indicated screw.
- Remove the belt retractor **(C)** complete with seat belt **(D)**, and replace both.
- Fit the new belt retractor **(C)** complete with seat belt **(D)**.
- Tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A

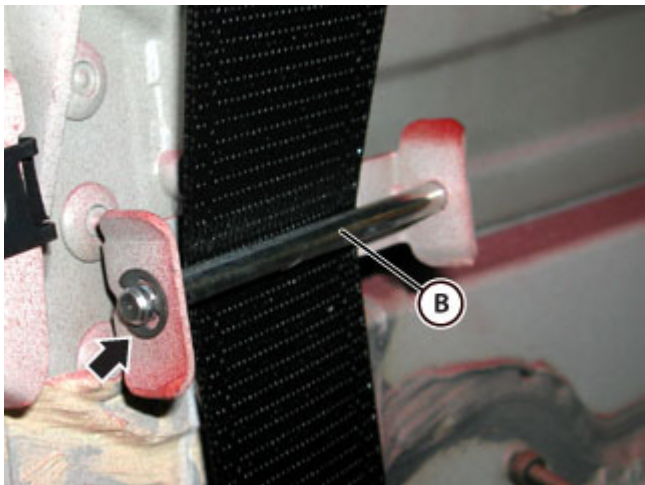


- Connect the connector (1).



- Fit the cover (E).
- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	4 Nm	B



- Fit the pin (B).
- Fit the indicated circlip.



- Fit the turning loop (A).
- Tighten the indicated screw.

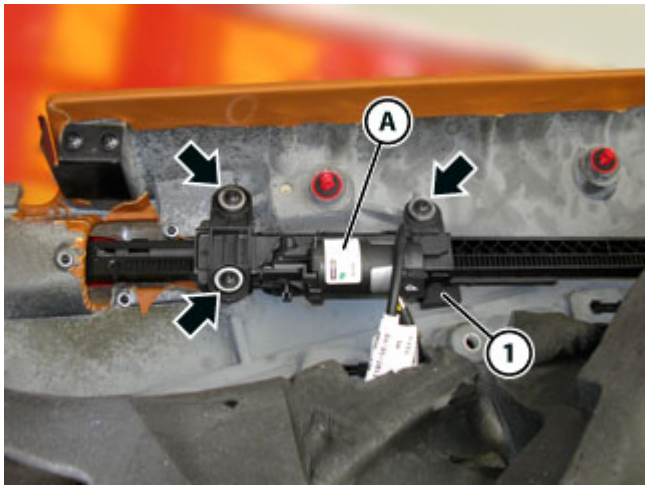
		
Tightening torque	Nm	Class
Screw	40 Nm	A

- Refit the roof trim panels (🔧 E4.04).
- Refit the upper B-pillar trim panels.
- Refit the passenger compartment trim panels (🔧 E4.04).
- Refit the lateral rear passenger compartment trim panels.
- Connect the battery (🔧 F2.01).
- Connect the DEIS tester to the diagnostic socket (🔧 F2.09).
- Perform the respective cycle with the DEIS diagnostic tester.

Replacing the front seat belt extenders

			
Tightening torque		Nm	Class
Fastening the front seat belt extenders	Screw	4 Nm	B

- Remove the passenger compartment trim panels (🔧 E4.04).
- Remove the lateral rear passenger compartment trim panels.



- Undo the screws indicated.
- Remove the seat belt extender **(A)** and replace.
- Replace the shim **(1)**.
- Remove the three fastener bushes from the new seat belt extender **(A)** and insert in the opposite side of the new seat belt extender **(A)**.
- Fit the new seat belt extender **(A)**.
- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	4 Nm	B

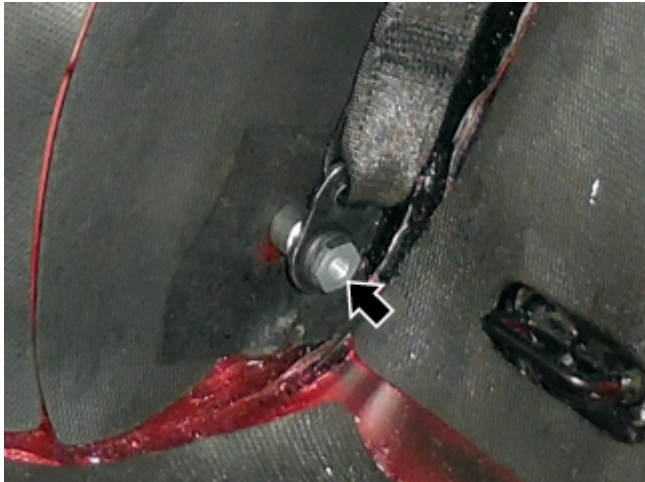
- Refit the passenger compartment trim panels ( E4.04).
-  Refit the lateral rear trim panels.

Replacing the rear seat belts

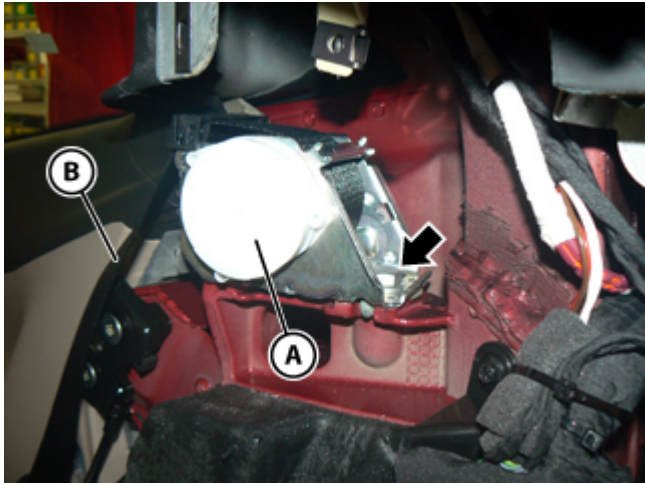
			
Tightening torque		Nm	Class
Fastening the rear seat belts	Screw	40 Nm	A
	Nut	40 Nm	A

- Remove the luggage compartment trim panels ( E4.06).
-  Remove the front lateral luggage compartment trim panels.
- Remove the rear seats ( E4.01).
-  Remove the seat cushions.

Right hand seat belt

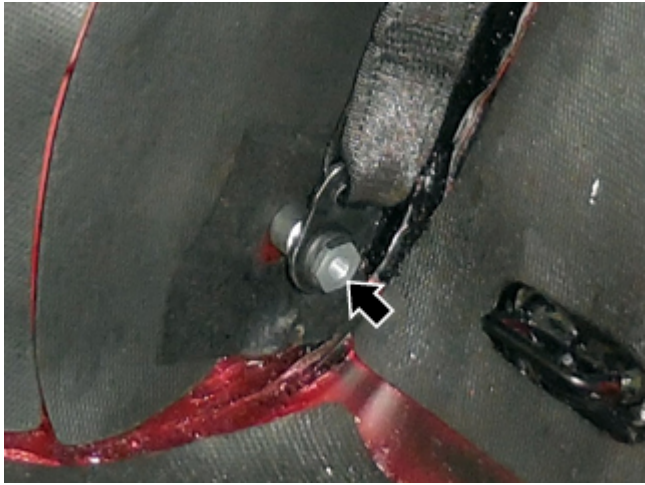


- Undo the indicated nut.



- Undo the indicated screw.
- Remove the belt retractor (A) complete with seat belt (B), and replace.
- Fit the new belt retractor (A), complete with seat belt (B).
- Tighten the indicated screw.

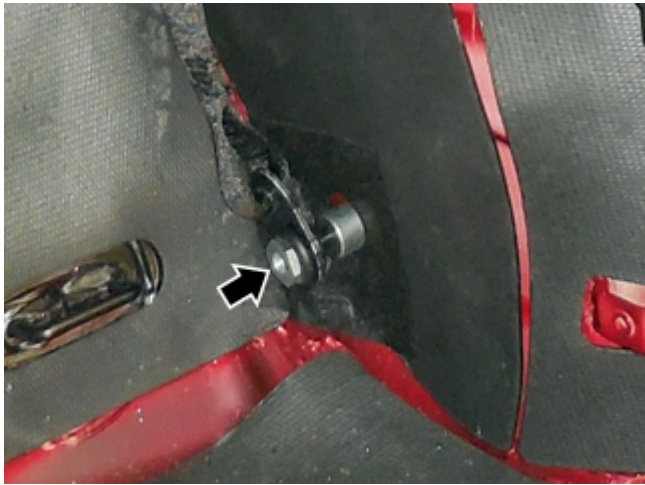
		
Tightening torque	Nm	Class
Screw	40 Nm	A



- Tighten the indicated nut.

		
Tightening torque	Nm	Class
Nut	40 Nm	A

Left hand seat belt



- Undo the indicated nut.



- Undo the indicated screw.
- Remove the belt retractor **(A)** complete with seat belt **(B)**, and replace.
- Fit the new belt retractor **(A)**, complete with seat belt **(B)**.
- Tighten the indicated screw.



Tightening torque	Nm	Class
Screw	40 Nm	A



- Tighten the indicated nut.



Tightening torque	Nm	Class
Nut	40 Nm	A

➤ Refit the rear seats ( E4.01).

 Refit the seat cushions.

➤ Refit the luggage compartment trim panels ( E4.06).

 Refit the front lateral luggage compartment trim panels.